Determinants

ANSWER KEY

2x2 determinants

1 Find the determinant of the matrix with rows $\langle 4,3\rangle, \langle 2,5\rangle$.

$4(5) - 3(2) = 20 - 6 = 14.$

2 Find the determinant of the matrix with rows $\langle -2,1\rangle, \langle 3,4\rangle$.

$-2(4) - 1(3) = -8 - 3 = -11.$

3x3 determinants via cofactor expansion

3 Find the determinant of the matrix with rows $\langle 1,2,3\rangle, \langle 0,1,4\rangle, \langle 5,6,0\rangle$ by cofactor expansion along the first row.

$1\det(\langle 1,4\rangle, \langle 6,0\rangle) - 2\det(\langle 0,4\rangle, \langle 5,0\rangle) + 3\det(\langle 0,1\rangle, \langle 5,6\rangle) = 1(0-24) - 2(0-20) + 3(0-5) = -24 + 40 -$

4 Find the determinant of the matrix with rows $\langle 2,0,1\rangle, \langle 3,1,0\rangle, \langle 1,2,4\rangle$ by cofactor expansion along the first row.

$2\det(\langle 1,0\rangle, \langle 2,4\rangle) - 0\det(\langle 3,0\rangle, \langle 1,4\rangle) + 1\det(\langle 3,1\rangle, \langle 1,2\rangle) = 2(4-0) - 0 + 1(6-1) = 8 + 5 = 13.$

5 Recompute the determinant of the matrix with rows $\langle 1,2,3\rangle, \langle 0,1,4\rangle, \langle 5,6,0\rangle$ by expanding along the SECOND row instead, to confirm the answer matches.

Expanding along row 2 (with alternating signs $-, +, -$): $-0\det(\langle 2,3\rangle, \langle 6,0\rangle) + 1\det(\langle 1,3\rangle, \langle 5,0\rangle) - 4\det(\langle 1,2\rangle, \langle 5,6\rangle) = 0 + 1(0-15) - 4(6-10) = -15 + 16 = 1$. Matches the earlier result of 1.

The determinant product rule

6 For A with rows $\langle 1,2\rangle, \langle 3,4\rangle$ and B with rows $\langle 2,1\rangle, \langle 0,3\rangle$, verify $\det(AB) = \det(A)\det(B)$.

$\det A = 4 - 6 = -2$, $\det B = 6 - 0 = 6$, so $\det A \det B = -12$. AB has rows $\langle 2,7\rangle, \langle 6,15\rangle$ (computed as usual matrix multiplication), and $\det(AB) = 2(15) - 7(6) = 30 - 42 = -12$. They match.

7 Using the product rule $\det(AB) = \det(A)\det(B)$, explain why $\det(A^{-1}) = \dfrac{1}{\det A}$ for an invertible matrix A.

Since $AA^{-1} = I$ and $\det I = 1$, the product rule gives $\det(A)\det(A^{-1}) = 1$. Dividing both sides by $\det A$ (nonzero since A is invertible) gives $\det(A^{-1}) = \dfrac{1}{\det A}$.

Determinants and scalar multiples / row operations

8 If A is a 2×2 matrix with $\det A = 5$, find $\det(2A)$ (scaling EVERY entry of A by 2).

For an $n \times n$ matrix, $\det(cA) = c^n \det A$. For $n = 2$: $\det(2A) = 2^2(5) = 20$.

9 If A is a 3×3 matrix with $\det A = 5$, find $\det(2A)$.

$\det(2A) = 2^3(5) = 40$ (note this is different from the 2×2 case — the exponent depends on the matrix's size).

10 Swapping two rows of a matrix negates its determinant. If A has rows $\langle 1, 2\rangle, \langle 3, 4\rangle$ with $\det A = -2$, find the determinant of the matrix with rows swapped: $\langle 3, 4\rangle, \langle 1, 2\rangle$.

Swapping rows negates the determinant: $\det = -(-2) = 2$. (Direct check: $3(2) - 4(1) = 6 - 4 = 2$.)

The invertibility criterion

11 State the invertibility criterion in terms of the determinant.

A square matrix A is invertible if and only if $\det A \neq 0$.

12 Is the matrix with rows $\langle 2, 4\rangle, \langle 1, 2\rangle$ invertible? Justify using the determinant.

$\det = 2(2) - 4(1) = 0$. Since the determinant is 0, the matrix is NOT invertible (singular).

13 Is the matrix with rows $\langle 1, 0, 2\rangle, \langle 0, 3, 1\rangle, \langle 2, 0, 4\rangle$ invertible? Justify using the determinant.

Expanding along column 2 (only one nonzero entry, 3, in position row 2):
$\det = -0(\ldots) + 3\det(\langle 1, 2\rangle, \langle 2, 4\rangle) - 0(\ldots) = 3(4 - 4) = 0$. Since $\det = 0$, the matrix is NOT invertible.

Applying the invertibility criterion

14 For what value(s) of k is the matrix with rows $\langle k, 2\rangle, \langle 3, k\rangle$ NOT invertible?

$\det = k^2 - 6$. Setting $k^2 - 6 = 0$ gives $k = \pm\sqrt{6}$. The matrix fails to be invertible exactly when $k = \sqrt{6}$ or $k = -\sqrt{6}$.

15 For what value(s) of k is the matrix with rows $\langle 1, k\rangle, \langle k, 4\rangle$ NOT invertible?

$\det = 4 - k^2$. Setting $4 - k^2 = 0$ gives $k = \pm 2$. The matrix is singular exactly when $k = 2$ or $k = -2$.

Geometric meaning of the determinant

16 For a 2×2 matrix A with rows $\mathbf{r}_1, \mathbf{r}_2$, $|\det A|$ gives the area of the parallelogram spanned by $\mathbf{r}_1$ and $\mathbf{r}_2$. Find this area for $\mathbf{r}_1 = \langle 3, 0\rangle$, $\mathbf{r}_2 = \langle 1, 4\rangle$.

$\det = 3(4) - 0(1) = 12$. Area $= |12| = 12$.

17 Explain, using the geometric meaning of the determinant, why a matrix with two identical rows always has determinant 0.

Two identical rows span a degenerate "parallelogram" with zero area (the two vectors are parallel, in fact equal, so they don't sweep out any two-dimensional region) — matching $\det = 0$.

Determinants of special matrices

18 Find the determinant of the identity matrix I_3 (rows $\langle 1,0,0\rangle, \langle 0,1,0\rangle, \langle 0,0,1\rangle$).

$\det I_3 = 1$ (expanding along any row/column, the product of the diagonal entries, all 1's, with every off-diagonal cofactor contributing 0).

19 Find the determinant of the matrix with rows $\langle 3,0,0\rangle, \langle 0,-2,0\rangle, \langle 0,0,5\rangle$ (a diagonal matrix).

For a diagonal (or more generally triangular) matrix, the determinant is the product of the diagonal entries: $3(-2)(5) = -30$.

20 Find the determinant of the matrix with rows $\langle 0,0,3\rangle, \langle 0,1,0\rangle, \langle 3,0,0\rangle$, and explain why it is nonzero even though the main diagonal has two zeros.

Expanding along row 2 (only nonzero entry is the middle 1): $\det = 1 \cdot \det(\langle 0,3\rangle, \langle 3,0\rangle) = 1(0-9) = -9$. The main-diagonal shortcut only applies to TRIANGULAR matrices; this matrix is not triangular (it has nonzero entries above AND below the diagonal), so the diagonal-product rule doesn't apply here.

21 Find the determinant of the zero matrix (all entries 0) of any size, and connect this to the invertibility criterion.

$\det = 0$ for the zero matrix of any size (every row is the zero vector, so every cofactor expansion term vanishes). This matches the invertibility criterion: the zero matrix is never invertible.

22 Explain why any matrix with an entire row (or column) of zeros must have determinant 0, using cofactor expansion along that row.

Expanding along a row of all zeros, every term in the cofactor expansion is (zero entry)$\times$(its cofactor)$=0$, so the whole sum is 0.

Cramer's Rule

23 Cramer's Rule solves $ax+by=e$, $cx+dy=f$ via $x = \frac{\det A_x}{\det A}$, $y = \frac{\det A_y}{\det A}$, where A is the coefficient matrix and A_x, A_y replace the respective column with $\langle e, f\rangle$. Use it to solve $2x+3y=8$, $x-y=1$.

A has rows $\langle 2,3\rangle, \langle 1,-1\rangle$, $\det A = -2-3 = -5$. A_x has rows $\langle 8,3\rangle, \langle 1,-1\rangle$, $\det A_x = -8-3 = -11$, so $x = \frac{-11}{-5} = \frac{11}{5}$. A_y has rows $\langle 2,8\rangle, \langle 1,1\rangle$, $\det A_y = 2-8 = -6$, so $y = \frac{-6}{-5} = \frac{6}{5}$.

24 Explain why Cramer's Rule fails (gives no unique solution) whenever $\det A = 0$.

Division by $\det A = 0$ is undefined, and this matches the invertibility criterion: $\det A = 0$ means the coefficient matrix is singular, so the system does not have a unique solution (it has either no solution or infinitely many).

25 Compare Cramer's Rule to Gaussian elimination as a solution method for large systems (say, 10 equations in 10 unknowns). Which is more practical, and why?

Gaussian elimination is far more practical for large systems: it requires roughly $O(n^3)$ operations, while Cramer's Rule requires computing $n+1$ separate $n \times n$ determinants, each costing far more via cofactor expansion — Cramer's Rule is mainly useful for small systems (2×2 or 3×3) or for theoretical formulas, not large-scale computation.